

Dhruvraj Singh Shekhawat

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Summary

B.Tech (Electrical and Communication Engineering) student at VIT Bhopal (Expected 2027) with hands-on work in simulation-driven engineering and machine learning pipelines. Strong in Python/C/C++, MATLAB, and model experimentation. Comfortable documenting experiments, analyzing performance metrics, and building modular scripts for training, evaluation, and testing.

Education

Vellore Institute of Technology, Bhopal

Expected 2027

B.Tech in Electrical and Communication Engineering

Coursework: Analog Electronics, Digital Circuits, Signal Processing, Control Systems, Communication Engineering, Embedded Systems

Universal Public School, Jaipur

2022

Senior Secondary

Stream: Physics, Chemistry, Mathematics

Technical Skills

Programming: Python, C, C++, Java

ML: Neural Networks, Model fine-tuning, Training/Evaluation Pipelines, NumPy, TensorFlow, scikit-learn

Hardware/Simulation: MATLAB/Simulink, LTspice, Circuit Analysis

Core: Algorithms, Mathematics, Signal Processing, IoT

Projects

Wireless Power Transfer System for Electric Vehicles

MATLAB, Ansys Maxwell

- Designed and simulated a dynamic wireless charging system; analyzed magnetic coupling, coil misalignment, and efficiency.
- Optimized coil spacing and geometry; improved simulated efficiency by **~15%** under tested conditions.
- Evaluated Maxwell coil designs; compared mutual inductance trends and alignment tolerance across configurations.

AASIST-Based Machine Learning Software

Python, TensorFlow

- Implemented a custom AASIST (Attention-Augmented Self-Improved Speech Transformer) training pipeline.
- Built preprocessing, training, and evaluation scripts using NumPy, TensorFlow, and scikit-learn.
- Improved accuracy by **8%** via hyperparameter tuning and experiment tracking/visualization.

Certifications

- MATLAB Onramp – Signal Processing & Digital Signal Processing (MathWorks)
- Introduction to Internet of Things – NPTEL (Elite, 86%)
- Introduction to Programming in C – NPTEL (Elite)
- Google IT Support - Google
- Applied Machine Learning in Python - University of Michigan

Achievements & Roles

- Presented a project on Wireless Power Transfer for Electric Vehicles at VIT Bhopal Project Exhibition.
- Achieved NPTEL Elite certification in C programming and Internet of Things (86%).
- Represented at the national level in Judo.